Proposal for Factory Financial Management & Role-Based Access Control Software

Total Proposed Development Cost: BDT 50,000
Estimated Development Period: 15–20 Working Days
Project Type: Integration and Enhancement of Existing Factory Management Software

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1. Project Overview

The proposed project aims to enhance the company's existing factory management software by developing and integrating two additional modules:

1. **Financial Management & Accounting System**
2. **Role-Based Access Control (RBAC) System**

The Financial Management module will digitize the company's existing Excel-based financial reporting process, automate financial calculations, and provide centralized access to financial information.

The Role-Based Access Control system will allow administrators to manage employee roles and permissions, ensuring that users can access only authorized modules and perform permitted operations.

Both modules will be integrated directly into the existing software, maintaining a unified dashboard, authentication system, and database architecture.

Note: The existing Raw Material Inventory and Mold Management modules have already been developed. These modules are excluded from the new development scope. However, relevant data from these modules may be used for financial calculations and reporting where applicable.

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2. Scope of Development

Module 1: Financial Management & Accounting System

Development Cost: BDT 30,000

The Financial Management module will be developed based on the company's supplied financial statement and existing accounting requirements.

A. Sales Management

- Record and manage monthly sales.
- Maintain separate sales categories:
    - Hanger
    - Loop
    - String
- Automatically calculate total sales.
- Generate monthly and yearly sales reports.
- Search and filter sales records.
- Edit and update sales information based on user permissions.

B. Raw Material Cost Management

- Record raw material expenses by category.
- Track raw material costs for Hanger, Loop, and String.
- Automatically calculate total raw material expenditures.
- Generate monthly raw material expense summaries.
- Integrate relevant existing raw material inventory data where applicable.

C. Factory Overhead Management

Record and manage factory-related expenses, including:

- Employee salaries, overtime, and benefits.
- Electricity and utility expenses.
- Machinery maintenance and repairs.
- Transportation and factory rent.
- Factory supplies and miscellaneous expenses.
- Commission expenses.
- Mold and machinery expenses.
- Depreciation expenses.

The system will calculate total factory overhead costs based on recorded expenses.

D. Administrative Expense Management

- Office employee salaries.
- Administrative transportation expenses.
- Commercial and administrative expenses.

- Other office operating expenses.
- Monthly administrative expense summaries.

E. Automated Financial Calculations

The system will automatically calculate the following financial values:

1. Total Revenue
2. Total Raw Material Cost
3. Total Factory Overhead
4. Total Production Cost
5. Gross Profit
6. Total Administrative Expenses
7. Net Profit
8. Net Profit Margin

All financial calculations will follow the agreed accounting structure based on the supplied financial statement.

F. Financial Dashboard & Reporting

- Monthly financial statements.
- Monthly and yearly sales summaries.
- Raw material cost summaries.
- Factory overhead reports.
- Administrative expense reports.
- Gross profit and net profit reports.
- Sales and expense comparison charts.
- Monthly financial performance charts.
- Filter financial reports by month and year.
- Export financial reports to Excel and PDF.

Module 2: Role-Based Access Control (RBAC)

Development Cost: BDT 20,000

A centralized Role-Based Access Control system will be integrated into the existing application to manage employee access across existing and newly developed modules.

A. Role Management

- Create, edit, and delete custom roles.
- Assign roles to employees.

- Define permissions for individual roles.
- Modify role permissions dynamically.
- Manage role assignments through an administrative dashboard.

B. Frontend Permission Management

- Control access to application dashboard pages.
- Dynamically manage sidebar and navigation menu visibility.
- Restrict access to unauthorized pages.
- Display only permitted modules and operations.
- Apply permissions to existing and newly developed software modules.

C. Backend API Authorization

- Implement server-side permission validation.
- Control View, Create, Update, and Delete operations.
- Protect application endpoints using authorization middleware.
- Reject unauthorized API requests.
- Prevent users from bypassing frontend permission restrictions.

D. Permission Management Dashboard

- Centralized permission-management interface.
- Module-wise permission selection.
- Bulk permission assignment.
- User-to-role assignment management.
- Protected system administrator role.
- Basic activity logging for critical permission changes.

E. Integration With Existing Modules

The RBAC system will support permission management for the existing application modules, including:

- Existing Raw Material Inventory Management.
- Existing Mold Management.
- New Financial Management & Accounting.
- Existing Dashboard and Administrative Features.

Permission enforcement will be implemented for the agreed existing pages, actions, and endpoints.

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3. Integration With Existing Software

Both modules will be developed and integrated into the company's current factory management application.

The integration will include:

- Reusing the existing authentication system where compatible.
- Integrating new functionality into the current dashboard and navigation.
- Connecting the Financial Management module with relevant existing data.
- Applying RBAC permissions to existing and newly developed modules.
- Maintaining compatibility with the current application database.
- Preserving existing inventory and mold management functionality.
- Testing the integrated application to minimize disruption to existing operations.

The client will not need to purchase or operate a separate software application.

All new features will be available within the existing factory management software.

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4. Security Features

The system will include:

- Secure user authentication.
- Role-based authorization.
- Backend API permission validation.
- Secure password storage.
- Server-side input validation.
- Protection against unauthorized data access.
- Prevention of unauthorized data modification.
- Basic administrative activity logs.

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5. Technical Implementation

The new modules will follow the existing application's technology stack and software architecture.

Component	Technology

Backend	Laravel
Frontend	React.js with Inertia.js
Database	MySQL
Architecture	Modular Application Structure
Authentication	Existing Authentication System
Authorization	Dynamic Role-Based Access Control
Server	Linux with Nginx and Docker, where compatible
Testing Environment	Developer-Hosted Subdomain
Final Deployment	Existing Client Production Environment

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6. Development Timeline

Estimated Completion: 15–20 Working Days

Development Phase	Estimated Duration
Existing system analysis and integration planning	1–2 Days
Financial Management module development	6–8 Days
RBAC system development	5–6 Days
Integration, testing, and bug fixing	3–4 Days

The development timeline begins after project confirmation, receipt of the initial payment, and availability of the necessary existing software access and requirements.

Additional time may be necessary if major changes to the existing system are identified or if client approval and feedback are delayed.

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7. Project Cost Breakdown

Development Component	Cost (BDT)

Financial Management & Accounting System	30,000
Role-Based Access Control (RBAC) System	20,000
Total Development Cost	**50,000**

Total Project Cost: BDT 50,000 (Fifty Thousand Taka Only)

The development cost includes:

- Development of both modules.
- Integration with the existing software.
- Implementation of agreed financial calculations.
- Frontend and backend RBAC implementation.
- Functional testing and bug fixing.
- Deployment to the testing environment.
- Final production migration following full payment.

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8. Payment Schedule

The project will follow a two-stage payment structure.

Payment Milestone	Amount (BDT)
Project Confirmation (Advance Payment)	10,000
After Testing & Approval, Before Production Migration	40,000
Total Project Payment	**50,000**

Payment Terms

Stage 1: Project Confirmation

The client will pay BDT 10,000 as an initial advance payment to confirm the project and initiate development.

Stage 2: Testing, Approval & Final Payment

Once development is completed, the software will be made available on a dedicated testing subdomain hosted on the developer's server.

The client will be able to test and verify the agreed functionality before making the final payment.

After successful testing and approval, the client will pay the remaining BDT 40,000.

The full project payment of BDT 50,000 must be completed before the software is migrated to the client's production environment.

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9. Testing, Approval & Production Migration

To ensure transparency and allow the client to verify the completed software, the following delivery process will be followed.

Step 1: Development

Both modules will be developed and integrated into a testing copy of the existing factory management application.

Step 2: Deployment to Testing Server

After development, the integrated software will be deployed to a dedicated subdomain on the developer's server.

The client will receive access credentials to test the application.

Step 3: Client Testing & Verification

The client will be able to:

- Verify monthly financial calculations.
- Review generated financial statements.
- Test financial dashboard features.
- Check sales and expense reporting.
- Test role creation and permission assignment.
- Verify frontend and backend access restrictions.
- Confirm integration with existing software functionality.

Any bugs related to the agreed development scope identified during testing will be corrected before final approval.

Step 4: Final Payment

Once the client has reviewed and approved the software in the testing environment, the remaining payment of BDT 40,000 must be completed.

Step 5: Production Migration

Following receipt of the full payment, the developer will migrate the completed modules into the client's existing production software environment.

The migration process will include:

- Database migrations and required configuration changes.
- Deployment of new application features.
- Integration verification.
- Basic production functionality testing.

Important: The testing environment is intended for evaluation and approval only. Final production migration and delivery will take place after receipt of 100% of the agreed project payment.

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10. Project Deliverables

Upon successful completion and final payment, the client will receive:

1. Fully integrated Financial Management & Accounting module.
2. Automated financial calculations and reporting.
3. Monthly and yearly financial dashboards.
4. Financial report export functionality in Excel and PDF.
5. Dynamic Role-Based Access Control system.
6. Centralized role and permission management dashboard.
7. Frontend and backend permission enforcement.
8. Integration with the existing software modules.
9. Tested and deployed application updates.
10. Basic administrator usage guidance.

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11. Project Terms & Conditions

1. **Existing Software Integration:** The new modules will be developed within the existing factory management software rather than as a standalone application.
2. **Development Scope:** Only Financial Management & Accounting and RBAC development are included in this proposal.

3. **Existing Modules:** Redevelopment of Raw Material Inventory and Mold Management modules is excluded. Only necessary and agreed integration work is included.
4. **Advance Payment:** An initial payment of BDT 10,000 is required before development begins.
5. **Testing Environment:** The completed software will be made available on a developer-hosted subdomain for client testing and verification.
6. **Final Payment:** The remaining BDT 40,000 must be paid after successful testing and approval, before production migration.
7. **Production Migration:** Migration to the client's production environment will be performed only after full project payment has been received.
8. **Additional Requirements:** Features or modifications outside the agreed project scope may require a separate quotation.
9. **Infrastructure Costs:** Domain, hosting, third-party subscriptions, and other infrastructure expenses are not included in the development cost.
10. **Existing Architecture:** Major restructuring of existing software, database architecture, or infrastructure is not included.
11. **Bug Fixing Support:** Bugs directly related to newly developed features will be corrected without additional development charges for 15 days following final production deployment.
12. **Project Timeline:** The estimated development duration is 15–20 working days, subject to timely access to the existing software and confirmation of requirements.

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12. Conclusion

The proposed development will enhance the company's existing factory management software with a comprehensive Financial Management & Accounting module and a centralized Role-Based Access Control system.

The new accounting functionality will reduce manual calculations, improve reporting accuracy, and provide better visibility into factory financial performance.

The RBAC system will strengthen application security and simplify the management of employee permissions across existing and newly developed modules.

By integrating both systems into the current software, the company can expand its operational capabilities without replacing its existing infrastructure or maintaining separate applications.

Project Summary

Financial Management & Accounting: BDT 30,000
Role-Based Access Control: BDT 20,000

Total Development Cost: BDT 50,000
Advance Payment: BDT 10,000
Remaining Payment: BDT 40,000
Estimated Development Period: 15–20 Working Days

The client will have the opportunity to test and approve the completed functionality on the developer's testing server before making the final payment.

Final production migration will be completed after full payment is received.

We look forward to supporting the continued development, security, and operational efficiency of the company's existing factory management software.